

Brandon Pardi

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Education

University of California, Davis

PhD in Computer Science (in progress)

September 2024 – June 2029

GPA 4.0

University of California, Merced

BS in Computer Science and Engineering (High Honors)

August 2021 – December 2023

GPA: 3.95

Fellowships/Honorships

- **CITRIS Innovation Fellowship** (Summer 2025)
- **US Census Bureau Data Science Fellow** (Summer 2024)
- **Outstanding Student Award for Computer Science and Engineering** (Fall 2023)
- **NSF-CREST Fellow** (Fall 2022 – Spring 2023)
- **Chancellor's Honor List** (Spring 2022)
- **Dean's Honor List** (Fall 2021, Fall 2022, Fall 2023)

Publications

- Pardi BM, Ahmed ST, Flores SJ, Maers LE, Yanez Soto B, Andresen Eguiluz RC. *pyQCM-BraTaDio: A tool for visualization, data mining, and modelling of Quartz crystal microbalance with dissipation data*. Journal of Open Source Software, 9(99), 6831 (2024). DOI: 10.21105/joss.06831
- Smith AM, Inocencio DG, Pardi BM, Gopinath A, Andresen Eguiluz RC. *Facile determination of the Poisson's ratio and Young's modulus of polyacrylamide gels and polydimethylsiloxane*. ACS Applied Polymer Materials 6(12): 2405-2416 (2024). DOI: 10.1021/acsapm.3c03154
- Pardi BM, Mummaneni G, Stephan Naunheim, Trigila C, Roncali E. *Gradient-Based Optimal Data Subset Selection Methods for reducing GAN training times*. 2025 (in preparation)
- Pardi BM, Adhikari A, Chandrasekhar S. *Improving Capstone Team Outcomes through Dynamic Skill Matching and Preference Alignment*. 2025 (in submission)

Experience

Graduate Student Researcher, Roncali Lab, UC Davis

October 2024 – Present, 30hrs/wk

- Developed novel methods for training GAN models with 75% less data while maintaining near-equivalent performance.
- Implemented more advanced statistical methods for evaluating GAN performance on tabular High Energy Physics data.
- Applied Bayesian inference with radial basis functions for hyperparameter optimization to optimize model performance.
- Contributing to open source development of GATE v10, a Python-based wrapper for the C++ particle physics simulator Geant4 from Cern labs.

Machine Learning Researcher, FOCAL Lab, UC Davis

June 2025 – August 2025, 30hrs/wk

- Implemented a fully configurable ViT transfer learning pipeline to achieve tree species image classification accuracy of 96%, and F1 score of 91%, an improvement upon lab's previous method results of 75% and 71% respectively.
- Applied gradient based data reduction methods to achieve near equivalent performance with 25% less data.

Computational Tool Developer, MECHANO3B[i]OLOGY Lab at UC Merced

June 2022 – June 2024, 20hrs/wk

- Developed scripts to process and analyze biophysics data in seconds, in lieu of manual spreadsheet work.
- Employed several computer vision tracking algorithms to analyze cell travel and deformation.
- Provided intuitive ways for researchers to navigate software.

Lawrence Livermore National Laboratory Data Science Challenge

July 2023, 40hrs/wk

- Classified different heart diseases from 12 lead ECG data.

- Designed a U-net like model architecture to predict activation graphs of 75 different points in the myocardium with an R2 score of 0.96.

Project Lead: Automated Rodent Detection Pipeline for Agrecom

January 2023 – May 2023, 20hrs/wk

- Assigned tasks for developing/integrating different facets of the program.
- Trained a custom image classifier to detect rodent presence from cameras with 91% accuracy.

Skills

Programming Languages

- Python (Advanced)
- SQL (Proficient)
- C/C++ (Proficient)
- HTML/CSS/JS (Proficient)

Soft Skills

- Self-motivated
- Independent
- Team-working
- Adaptability

Technical Skills

- Data Science/Machine Learning
- Temporal & Spatial Modeling
- Mathematics & Statistics
- Classification, Regression & Generative Models
- Computer Vision
- Software Development & Algorithm Design
- LLMs & Agent Development
- Database Systems

Domain Expertise

- Biophysics
- Biomedical
- High Energy Physics
- Climatology
- Surface Sciences
- Cell Mechanics
- Educational Technology
- Civics and Public Administration

Contactable References Available Upon Request